Concentric-Ring Electrode Systems for Noninvasive Detection of Single Motor Unit Activity

Dario Farina, *Member, IEEE,* and Corrado Cescon

***Abstract*—New recording techniques for detecting surface electromyographic (EMG) signals based on concentric-ring electrodes are proposed in this paper. A theoretical study of the two-dimensional (2-D) spatial transfer function of these recording systems is developed both in case of rings with a physical dimension and in case of line rings. Design criteria for the proposed systems are presented in relation to spatial selectivity. It is shown that, given the radii of the rings, the weights of the spatial filter can be selected in order to improve the rejection of low spatial frequencies, thus increasing spatial selectivity. The theoretical transfer functions of concentric systems are obtained and compared with those of other detection systems. Signals detected with the ring electrodes and with traditional one-dimensional and 2-D systems are compared.**

The concentric-ring systems show higher spatial selectivity with respect to the traditional detection systems and reduce the problem of electrode location since they are invariant to rotations. The results shown are very promising for the noninvasive detection of single motor unit (MU) activities and decomposition of the surface EMG signal into the constituent MU action potential trains.

***Index Terms*—Electromyography, multiple-electrode systems, spatial filters, spatial selectivity, surface electrodes.**

I. Introduction

COMPARED to invasive techniques for recording the electromyogram (EMG), the surface recording has the advantages of 1) avoiding discomfort and the possibility of infections and 2) providing an indication of the muscle activity in a wide region of space. The latter characteristic has been indicated since the pioneering works of Bouisset and Maton [3] as a fundamental advantage with respect to the very selective wire/needle recordings. Nevertheless many neuromuscular disorders can be investigated only by studying the activities of the single motor units (MUs) which is difficult if surface EMG is used. Invasive recordings have made possible the investigation of a number of basic physiological issues [5], [6] and have lead to the development of clinical applications [39]. Large research efforts on surface EMG focused in the past on the development of global processing techniques without aiming at the separation of the activities of individual MUs. Investigation of single MU activity from the surface EMG signal was based on different approaches, the most prevalent ones being spatial filtering and spatial sampling. One-dimensional (1-D) spatial filtering techniques (with electrodes placed along the muscle fibers) have been proposed, formally described, and applied by Broman [4], De Luca [26], Lindstrom [24], Masuda [25], and Merletti [26] and co-workers. The single-differential filter constitutes by far the most widely used 1-D detection system. Gydikov and co-workers [19] introduced the superficial branched electrodes, the simplest one being constituted by a single pole in the middle and two short-circuited leading-off areas symmetrically placed on both sides. Although this system is used also in the direction of the muscle fibers providing, as theoretically predicted, selectivity performance similar to that of the double-differential filter [22], the best position of such an electrode was found to be perpendicular to the muscle fibers [19], [23]. In more recent years, Rau and co-workers proposed and developed the theoretical background for two-dimensional (2-D) systems constituted by point electrodes forming regular grids [34], [35]. These systems have proven to be powerful tools for the analysis of single MU properties [9], [31]–[33]. The basic ideas introduced in [34] and [35] have been more recently applied to investigate new strategies for 2-D grid spatial filter design [8], [11]. The detection of spatially filtered surface EMG signals from different points in space has opened remarkable possibilities such as the estimation of conduction velocity of single MUs [12], [14], [28], [32], the detection of anatomical characteristics of MUs [25], [27], [36]–[38], and the classification of single MU action potentials (MUAPs) as belonging to different MUs [10], [17], [18], [21], [27]. Systems of up to one hundred electrodes have been used [21], [30].

The spatial filters proposed in the past are based on the weighted summation of signals detected by a number of equally spaced electrodes. Recently, it has been shown [7], [13], [16], [20] that the physical dimensions of the recording electrodes can be modeled with a spatial filtering operation. In previous works [13], [16], we have derived the equivalent transfer functions of a number of electrode shapes indicating that electrode size performs a spatial filtering of the detected potential of the same type as that performed by the weighted summation of signals detected in different points in space.

It is the purpose of this paper to propose new spatial filters for surface EMG detection based on multiple concentric-ring electrodes. Kossev and co-workers [23] tested a system made of a central pole and a single concentric ring and described it assuming the averaging of the potential distribution under the ring area. Bhullar and co-workers [2] more recently also proposed a point electrode surrounded by a ring electrode with opposite and balanced weights and introduced a 1-D spatial frequency

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*D. Farina is with the Department of Electronics, Politecnico di Torino, Corso Duca degli Abruzzi 24, Torino, 10129, Italy, and also with the Dépt. d'Automatique et Informatique appliquée, Ecole Centrale de Nantes, Nantes, France (e-mail: farina@athena.polito.it).

C. Cescon is with the Department of Electronics, Politecnico di Torino, Torino, 10129, Italy.

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approach for its theoretical description. These past works are the bases for the present work whose objective is to go forward with the theoretical description of concentric-ring electrodes extending the theory to the 2-D spatial frequency domain and to multiple-ring systems.

II. Methods

A. Spatial Integration Performed by an Electrode With Physical Dimensions

Given a spatial distribution of an electric potential $\phi(x,z)$ in the (x,z) skin plane, under the approximation of negligible potential field distorsion by the presence of the electrode [40], an electrode with physical dimensions detects the spatial average of the potential distribution under the electrode area [13], [20]. We showed that the average of the 2-D potential distribution can be viewed as a 2-D filtering operation [13]. The 2-D spatial impulse response of the filter describing electrode shape is [13]

$$h_{\text{size}}(x,z) = \begin{cases} \frac{1}{S} & f(-x,-z) \le 0 \\ 0 & f(-x,-z) > 0 \end{cases} \tag{1}$$

where $f(x,z)$ is the function describing the electrode shape and S the electrode area.

The 2-D spatial transfer function to be applied to the potential distribution in case of a nonpoint electrode is the 2-D Fourier transform of the impulse response (1).

B. Equivalent 2-D Transfer Function of Concentric-Ring Electrode Systems

From [13] and [16], the equivalent transfer function of a circular electrode of radius r is given by

$$H_{\text{circ}}(k_x,k_z) = \begin{cases} 2\frac{J_1(rk_y)}{rk_y} & k_y \neq 0 \\ 1 & k_y = 0 \end{cases} \tag{2}$$

with $h_{\text{size}}(x,z) = 1/(\pi r^2)$ for $x^2+z^2 \le r^2$ and $h_{\text{size}}(x,z) = 0$ for $x^2+z^2 > r^2$, $k_x = 2\pi f_x$ and $k_z = 2\pi f_z$ the spatial angular frequencies in the two spatial directions and $J_1(z)$ the Bessel function of the first kind of order one [1].

From (1) and (2) and basic integral properties, we obtain the equivalent transfer function of a circular ring electrode

$$\begin{aligned} H_{\text{ring}}(k_x,k_z) &= \iint_{C_0} e^{-jk_x x} e^{-jk_z z} \mathrm{d}x\mathrm{d}z \\ &= \iint_{C_2} e^{-jk_x x} e^{-jk_z z} \mathrm{d}x\mathrm{d}z \\ &\quad - \iint_{C_1} e^{-jk_x x} e^{-jk_z z} \mathrm{d}x\mathrm{d}z \end{aligned} \tag{3}$$

with $C_0 = \{(x,z) : r_1^2 \le x^2+z^2 \le r_2^2\}$, $C_1 = \{(x,z) : x^2+z^2 \le r_1^2\}$, $C_2 = \{(x,z) : x^2+z^2 \le r_2^2\}$ and r_2 and r_1 the external and internal radii of the ring, respectively.

From (2) and (3), we obtain

$$H_{\text{ring}}(k_x,k_z) = \begin{cases} \frac{2}{(r_2^2-r_1^2)}\left[r_2\frac{J_1(r_2k_y)}{k_y} - r_1\frac{J_1(r_1k_y)}{k_y}\right] & k_y \neq 0 \\ 1 & k_y = 0. \end{cases} \tag{4}$$

A system which detects the weighted summation of the signals recorded by a circular electrode and M electrode rings with centers coincident with that of the circular electrode has the equivalent transfer function (see (5) at the bottom of the page) with $\{r_h\}_{h=0}^{2M}$ the radii of the electrodes (r_0 the radius of the circular electrode, r_1 and r_2 the internal and external radius of the first ring and so on) and $\{K_h\}_{h=0}^{M}$ the weights given to the $M+1$ detection surfaces. The transfer function (5) depends on the physical properties of the system (radii and thickness of the rings) and on the weights chosen. It is possible to build rings with small thickness and a small central electrode. In this way, the central electrode can be considered as a point electrode and the rings as circumferences. The spatial transfer function depends in this case on the radii of the rings and on the weights. With this approximation the effect of electrode size is neglected in the same way as it is done for the derivation of the traditional spatial filter transfer functions with the assumption of point electrodes [35].

From the analytical point of view, the transfer function of a ring with negligible thickness is obtained from (4) with a limit operation. In particular, by setting

$$r_2 = \overline{r} + \frac{\Delta r}{2} \quad r_1 = \overline{r} - \frac{\Delta r}{2}$$

with $\overline{r}$ the average radius of the ring and Δr its thickness, we can write (4) as

$$H_{\text{ring}}(k_y) = \begin{cases} \frac{1}{\overline{r}\Delta r}\left[(\overline{r}+\frac{\Delta r}{2})\frac{J_1((\overline{r}+\frac{\Delta r}{2})k_y)}{k_y} \right. \\ \quad \left. - (\overline{r}-\frac{\Delta r}{2})\frac{J_1((\overline{r}-\frac{\Delta r}{2})k_y)}{k_y}\right] & k_y \neq 0 \\ 1 & k_y = 0. \end{cases} \tag{6}$$

For $k_y \neq 0$, the limit for $\Delta r \longrightarrow 0^+$ leads to

$$\begin{aligned} H_{\text{appr}}(k_y) &= \lim_{\Delta r \longrightarrow 0^+} \left\{ \frac{J_1\left[\left(\overline{r}+\frac{\Delta r}{2}\right)k_y\right] - J_1\left[\left(\overline{r}-\frac{\Delta r}{2}\right)k_y\right]}{\Delta r k_y} \right. \\ &\qquad \left. + \frac{J_1\left[\left(\overline{r}+\frac{\Delta r}{2}\right)k_y\right] + J_1\left[\left(\overline{r}-\frac{\Delta r}{2}\right)k_y\right]}{2\overline{r}k_y} \right\} \\ &= \frac{\mathrm{d}J_1(u)}{\mathrm{d}u}\Big|_{u=\overline{r}k_y} + \frac{J_1(\overline{r}k_y)}{\overline{r}k_y}. \end{aligned} \tag{7}$$

Since for any complex u we have [1]:

$$uJ_n'(u) = nJ_n(u) - uJ_{n+1}(u) = -nJ_n(u) + uJ_{n-1}(u) \tag{8}$$

$$H_{\text{Mring}}(k_x,k_z) = \begin{cases} K_0\frac{2}{r_0}\frac{J_1(r_0k_y)}{k_y} + \sum_{i=1}^{M} K_i \\ \quad \times \left\{\frac{2}{(r_{2i}^2-r_{2i-1}^2)}\left[r_{2i}\frac{J_1(r_{2i}k_y)}{k_y} - r_{2i-1}\frac{J_1(r_{2i-1}k_y)}{k_y}\right]\right\} & k_y \neq 0 \\ \sum_{i=0}^{M} K_i & k_y = 0 \end{cases} \tag{5}$$

we obtain

$$H_{\mathrm{appr}}(k_y) = J_0(\overline{r}k_y) \tag{9}$$

which is the equivalent transfer function of a ring electrode with negligible thickness (limit of (6) for $\Delta r \to 0^+$). Transfer function (9) corresponds to a low pass filter whose cut-off frequency depends on the radius of the ring (it decreases with increasing $\overline{r}$).

Note that the same result can be obtained assuming negligible thickness for determining the impulse response and by computing the curvilinear integral:

$$H_{\mathrm{appr}}(k_y) = \frac{1}{2\pi\overline{r}} \int_\gamma e^{-jk_x x} e^{-jk_z z} \tag{10}$$

where

$$\gamma = (\overline{r}\cos t, \overline{r}\sin t,) \quad t \in [0, 2\pi)$$

is the curve which describes the ring shape. Equation (10) can be interpreted as the Fourier transform of the impulse response

$$h_{\mathrm{appr}}(x,z) = \frac{1}{2\pi\overline{r}} \delta(x^2 + z^2 - \overline{r}^2) \tag{11}$$

which is the impulse response of a ring with zero thickness. From the physical point of view, the convolution of the impulse response (11) with a 2-D spatial potential distribution is in any point (x_0, z_0) the average of the potential distribution along the circumference $(x - x_0)^2 + (z - z_0)^2 = \overline{r}^2$.

Observing that $||\gamma'|| = \overline{r}$, we obtain from (10)

$$H_{\mathrm{appr}}(k_y) = \frac{1}{2\pi} \int_0^{2\pi} e^{-jk_x \overline{r}\cos t} e^{-jk_z \overline{r}\sin t} dt. \tag{12}$$

The substitution

$$\begin{cases} k_x = k_y \cos\eta \\ k_z = k_y \sin\eta \end{cases}$$

and the definition of the Bessel functions of the first kind [1] lead from (12) to

$$\begin{aligned} H_{\mathrm{appr}}(k_y) =& \frac{1}{2\pi} \int_0^{2\pi} e^{-jk_y \overline{r}\cos(\eta - t)} dt \\ =& \frac{1}{\pi} \int_0^{\pi} e^{-jk_y \overline{r}\cos t} dt = J_0(\overline{r}k_y) \end{aligned}$$

which is the same result given by (9).

From (9) and assuming the central electrode to be a point electrode, it is possible to obtain the extension of (5) to rings with negligible thickness

$$H_{\mathrm{Mappr}}(k_y) = K_0 + \sum_{i=1}^{M} K_i J_0(r_i k_y) \tag{13}$$

with $\{r_i\}_{i=1}^M$ the radii of the ring electrodes and $\{K_i\}_{i=0}^M$ the weights given to the $M+1$ detection surfaces.

Equation (13) contains $2M+1$ parameters (the radii and the weights) which can be chosen in order to set some properties of the resultant transfer function.

C. Spatial Selectivity and Spatial Filter Design

Approximation (13) will be used to derive design criteria for the concentric systems. Removal of the 2-D spatial dc component from the detected signal leads to a first condition on the weights, independently of the radii

$$H_{\mathrm{Mappr}}(0,0) = 0 \implies \sum_{i=0}^{M} K_i = 0 \tag{14}$$

since $J_0(0) = 1$. Note that this condition is valid also in case of rings with nonzero thickness [see (5)].

By setting condition (14), the system is an isotropic 2-D filter with a first-order zero in $(k_x, k_z) = (0,0)$. If we impose a zero of order n at dc, the selectivity will increase. To do so it is sufficient to compute the first n derivatives of the transfer function (13) and set them to zero for $k_y = 0$.

The first derivative of (13) is computed by using (8)

$$\frac{\mathrm{d}H_{\mathrm{Mappr}}(k_y)}{\mathrm{d}k_y} = -\sum_{i=1}^{M} r_i K_i J_1(r_i k_y). \tag{15}$$

We obtain:

$$\frac{\mathrm{d}H_{\mathrm{Mappr}}}{\mathrm{d}k_y}(0) = 0$$

for any choice of the radii and weights.

Similarly, the second derivative of (13) is computed from (15)

$$\frac{\mathrm{d}^2 H_{\mathrm{Mappr}}(k_y)}{\mathrm{d}k_y^2} = -\sum_{i=1}^{M} r_i^2 K_i \left[J_0(r_i k_y) - \frac{J_1(r_i k_y)}{r_i k_y} \right]. \tag{16}$$

Setting it to zero for $k_y = 0$ leads to

$$-\sum_{i=1}^{M} r_i^2 K_i = 0.$$

The computation of the derivatives of higher order can be done again by using (8). We obtain that the condition

$$\frac{\mathrm{d}^n H_{\mathrm{Mappr}}}{\mathrm{d}k_y^n}(0) = 0 \tag{17}$$

is satisfied for any choice of the radii and weights if n is odd. To null the derivatives of even order the weights are obtained by solving the following linear system in $M-1$ equations (to be considered together with (14))

$$\underline{\underline{\mathbf{A}}}(r_1, \ldots, r_M) \cdot \underline{\mathbf{K}} = -\underline{\mathbf{1}} \quad \underline{\mathbf{K}} = -\underline{\underline{\mathbf{A}}}(r_1, \ldots, r_M)^{-1} \cdot \underline{\mathbf{1}} \tag{18}$$

with

$$\underline{\underline{\mathbf{A}}}(r_1, \ldots, r_M) = \begin{pmatrix} \left(\frac{r_2}{r_1}\right)^2 & \left(\frac{r_3}{r_1}\right)^2 & \cdots & \left(\frac{r_M}{r_1}\right)^2 \\ \left(\frac{r_2}{r_1}\right)^4 & \left(\frac{r_3}{r_1}\right)^4 & \cdots & \left(\frac{r_M}{r_1}\right)^4 \\ \vdots & \ddots & \ddots & \vdots \\ \left(\frac{r_2}{r_1}\right)^{2(M-1)} & \left(\frac{r_3}{r_1}\right)^{2(M-1)} & \cdots & \left(\frac{r_M}{r_1}\right)^{2(M-1)} \end{pmatrix}$$

and

$$\underline{\mathbf{K}} = \begin{pmatrix} \frac{K_2}{K_1} \\ \frac{K_3}{K_1} \\ \vdots \\ \frac{K_M}{K_1} \end{pmatrix} \quad \underline{\mathbf{1}} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}.$$

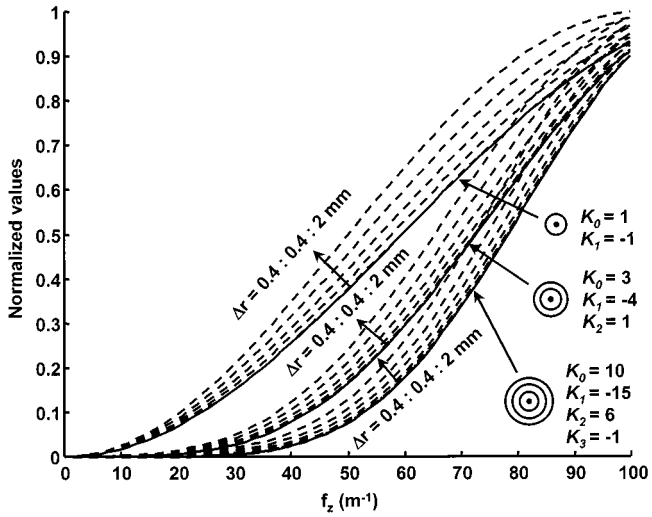

Fig. 1. Sections of the magnitude of the transfer functions of three concentric-ring systems whose weights have been designed in order to null the derivatives at $k_y = 0$ until the order $2M-1$, with M the number of concentric rings (radii multiple of 5 mm). The sections corresponding to approximated transfer functions (line rings and point central electrode) and to the case of nonzero thickness of the rings and circular central electrode (the diameter of the central electrode is equal to the thickness of the rings) are shown (thickness changing from 0.4 to 2 mm in steps of 0.4 mm). The approximation of line rings leads to negligible changes for thickness smaller than 1 mm. Each section has been normalized with respect to its maximum value for comparison.

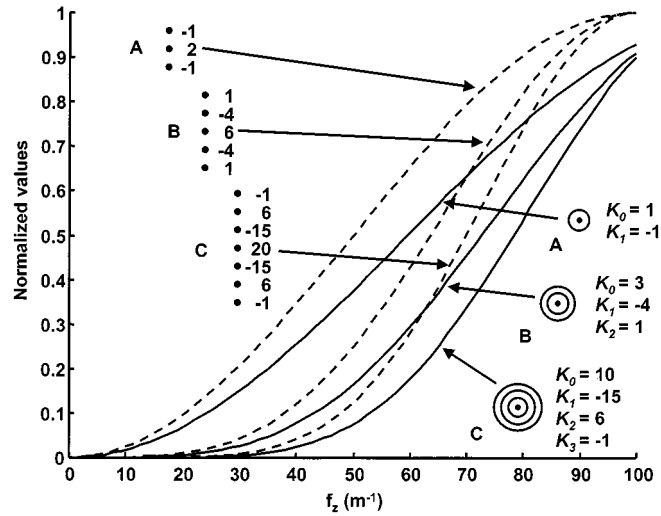

Fig. 2. Comparison between the sections in z direction of the magnitude of the transfer functions of three concentric-ring systems (solid lines) and three linear arrays implementing differential filters (dashed lines). Radii multiple of 5 mm, interelectrode distance 5 mm. Note the higher spatial selectivity of the concentric systems. The sections have been normalized with respect to their maximum values for comparison. The spatial extension in the z direction is the same for systems A, B, and C, respectively.

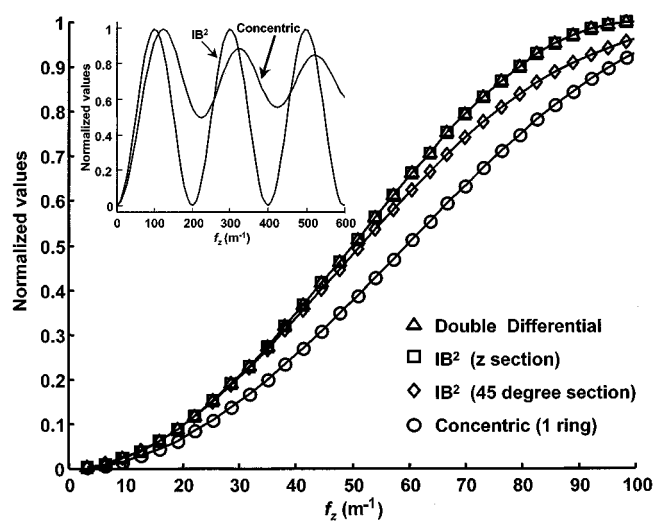

Fig. 3. Comparison between the magnitude of the transfer functions of the one-ring concentric system (section), of the 1-D double-differential filter (section in z direction) and of the inverse binomial filter of order two (sections in z direction and at 45°). Radii multiple of 5 mm, interelectrode distance 5 mm. Note the periodicity of the transfer functions of systems of equally spaced point electrodes (inset plot at upper left). The comparison between sections of the transfer function of the concentric system and of the inverse binomial filter of order two for high frequencies is also shown. The sections have been normalized with respect to their maximum values for comparison.

It is easy to show that the system (18) admits a unique solution in the condition $r_1 \neq r_2 \neq \cdots \neq r_{M-1} \neq r_M$.

System (18) provides a closed form for computing the weights given the number of rings and the radii in order to satisfy condition (17) for $n = 1, \ldots, 2M-1$. Solving the system (18) leads to the conditions of null derivatives at $k_y = 0$ until order $2M-1$. Condition (14) determines the value of K_0. It is evident that the conditions (14) and (18) lead to the same weights if all the radii are multiplied by a common factor.

Fig. 1 shows a section of the transfer functions of concentric-ring systems with weights obtained by (14) and (18). The improvement in rejection of low frequencies as the number of rings increases is evident. The figure also reports the comparison between the approximated and the exact transfer function in the case of different thickness and number of rings. Condition (14) is satisfied also in case of nonzero ring thickness while condition (17) is only approximated when the rings have a physical thickness.

D. Comparison of Concentric-Ring System With Traditional Recording System Transfer Functions

Fig. 2 shows the comparison between the sections in z direction of the transfer functions of three concentric-ring systems with weights assigned with criteria (14) and (18) and three 1-D differential filters. Spatial selectivity increases with increasing differentiation order and the number of rings. Concentric-ring systems are more selective with respect to traditional 1-D differential systems in the z section for equal extension of the recording system in the z direction. Moreover, concentric-ring systems show the same characteristics in any direction and, as a consequence, are not sensitive to fiber orientation.

Fig. 3 shows the comparison between the transfer functions of the one-ring concentric system (one point electrode and one ring), the 1-D double-differential filter and the second-order inverse binomial filter [8]. The concentric-ring system is invariant to rotations, thus, just one of its sections is reported; the 1-D double-differential filter is constant in the transversal dimension and the inverse binomial filter transfer function has the same sections in the z and x directions. Representative sections of these three detection system transfer functions are shown in the figure. The section of the 1-D double-differential filter transfer function is theoretically equivalent to the sections in x and z direction, respectively, of the basic branched electrode placed either perpendicular or parallel to the muscle fibers (as-

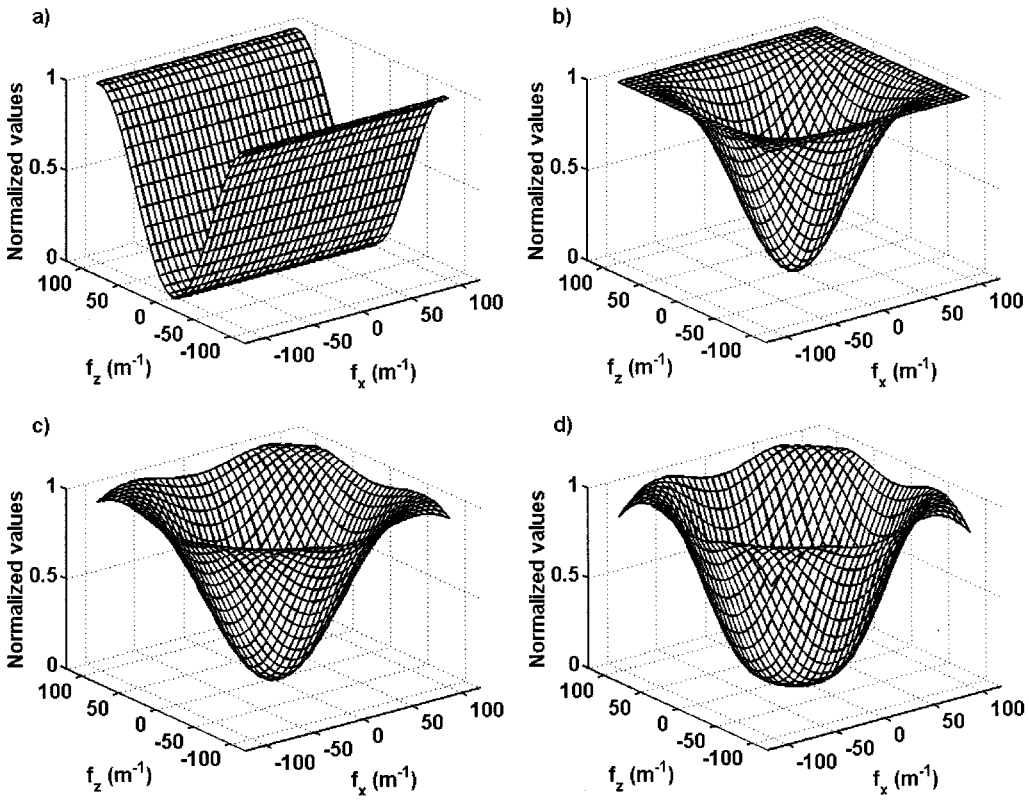

Fig. 4. magnitude of the 2-D transfer functions of (a) the double-differential system, (b) the inverse binomial filter of order two, (c) the concentric one-ring, and (d) the concentric two-ring system. Radii multiple of 5 mm, interelectrode distance 5 mm. The transfer functions have been normalized with respect to their maximum values for comparison.

suming identical contact impedences of the electrodes). One-dimensional double-differential filter is also equivalent to inverse binomial of order two in the z direction as it is obtained analytically. The one-ring concentric system shows the theoretically highest rejection of low spatial frequencies among the three systems. Note that the transfer functions of systems constituted by equally spaced point electrodes are periodic with period equal to the inverse of the interelectrode distance since they constitute 2-D finite impulse response (FIR) filters in the spatial domain [33]. The transfer functions of concentric systems are not periodic, thus, such systems act as high pass analog filters.

Fig. 4 shows in the two spatial dimensions the transfer functions of the double-differential filter [Fig. 4(a)], the inverse binomial of order two[Fig. 4(b)], and the one- [Fig. 4(c)],and two-ring [Fig. 4(d)] concentric systems. The 1-D double-differential filter is highly anisotropic (it depends only on one spatial direction), the inverse binomial filter of order two is approximately isotropic only for small wave numbers, while the two concentric systems are isotropic. It is evident the improvement in theoretical spatial selectivity of the two-ring concentric system with respect to the one-ring system.

III. Model Based and Experimental Validation of the Concentric Ring Systems

A. Simulation Methods and Recording Modalities

A mathematical model which allows the simulation of MUAPs generated by finite-length muscle fibers and detected by spatial filters with electrodes with physical dimensions has been previously developed [13]. The transfer function of the concentric systems (13) has been included in the model in order to compare the new detection systems with those proposed in the past. The three-pole branched electrode system with electrodes perpendicular to the muscle fibers has also been simulated in ideal conditions including in the model the transfer function of the double-differential system in the direction perpendicular to the muscle fibers.

To compare the different recording systems on the basis of experimental signals, an electrode system constituted by a 3×3 matrix of silver-point electrodes and a two concentric-ring detection system (silver electrode surfaces) was built. The two detection systems were next to each other on the same support. The thickness of the rings was 1 mm. Voltage followers were mounted near the electrodes. The interelectrode distance of the matrices of electrodes was 5 mm (equal in the two directions), while the radii of the two concentric rings were 5 and 10 mm. A second recording system constituted by a point electrode and five rings with radii multiple of 2.5 mm has also been developed. Surface EMG signals were collected with a 16 channel amplifier (LISiN-SEMA Elettronica, Torino, Italy), bandpass filtered (3-dB bandwidth 10–500 Hz), sampled at a rate of 2048 Hz and stored on a PC after 12-bit analog-to-digital conversion. Experimental EMG signals were recorded from the biceps brachii, the tibialis anterior and the abductor digiti minimi muscle. Before detecting spatially filtered signals, the muscle was scanned by a 16-electrode linear array [27] with 2.5-mm interelectrode distance. The aim was to identify the innervation zones and tendon

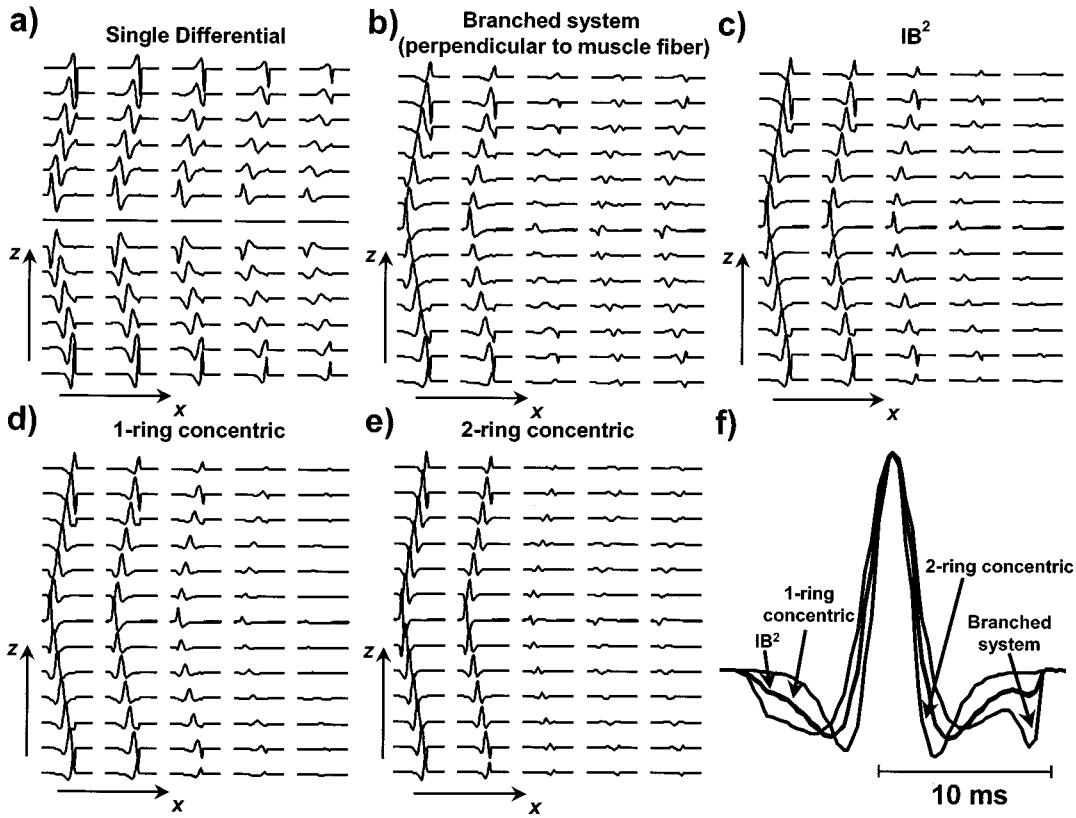

Fig. 5. (a)–(e) Simulated signals detected along a muscle fiber at different transversal distances from the fiber by five detection systems . The fiber is at 2-mm depth in the muscle, the fat and skin layers are 3 and 1 mm thick, respectively, the fiber semi-lengths are 60 mm in both directions and the recordings in x directions are at distances from 0 mm (detection systems on the fiber) to 10 mm at steps of 2.5 mm. The interelectrode distance in (a)–(c) is 5 mm, while the radii in (d) and (e) are multiple of 5 mm. In each case, 13 electrode system locations along the fiber (z direction) have been simulated. The distance between each location along the fiber is 10 mm. (f) A comparison of the duration of the surface detected potentials is also shown. In (f), the potentials are detected for each system at a transversal distance of 2.5 mm from the fiber, in the middle between the innervation zone and tendon region.

regions in order to place the detection systems in the region corresponding to unidirectionally propagating signals. The spatial filter output signals were obtained by linear combination of the signals detected by the different detection surfaces. A three-pole branched system with electrodes perpendicular to the muscle fibers has also been compared with the other systems in ideal conditions, that is combining the detected signals in order to obtain a double-differential recording transversal to fiber direction.

Representative results from the simulations and the collected signals are presented in the following.

B. Comparison of EMG Signals Detected by Different Recording Techniques

Fig. 5 shows simulated MUAPs detected with different spatial filters along a muscle fiber at different transversal distances from the source. It appears that the single-differential detection system is the least selective one in transversal direction (Fig. 5(a)) while the branched electrode with poles transversal to the muscle fibers shows high selectivity in transversal direction (comparable to that of the concentric system with two rings) [Fig. 5(b)]. However the branched electrode shows lower selectivity in the longitudinal direction with respect to all the systems except the single differential [Fig. 5(f)]. This is mainly due to the fact that this recording system has a main direction of selectivity. The concentric system of order two shows the best performance.

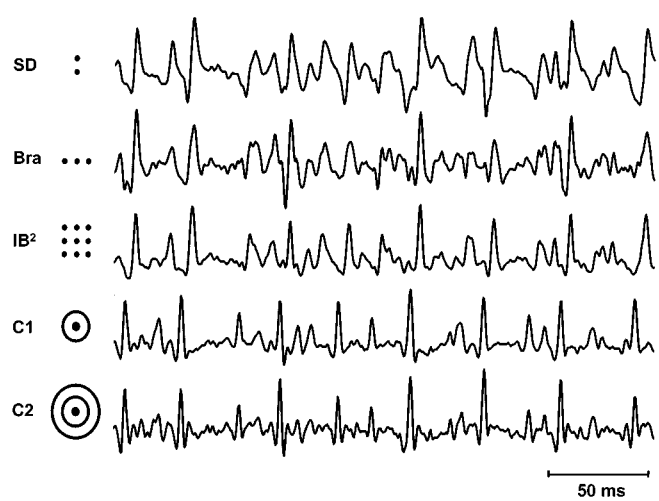

Fig. 6. Surface EMG signals simultaneously recorded with different spatial filters. Biceps brachii muscle, 30% MVC contraction level. Single-differential, branched system transversal to the muscle fibers, inverse binomial filter of order two, and concentric one- and two-ring signals are shown. The concentric system is closer to the innervation zone with respect to the grid systems as it can be observed from the time delay separating the signals detected by the two concentric systems and those detected by the single differential, the branched and the inverse binomial of order two systems.

Fig. 6 shows 300-ms-long segments of experimental signals obtained by linear combinations of the signals detected by the

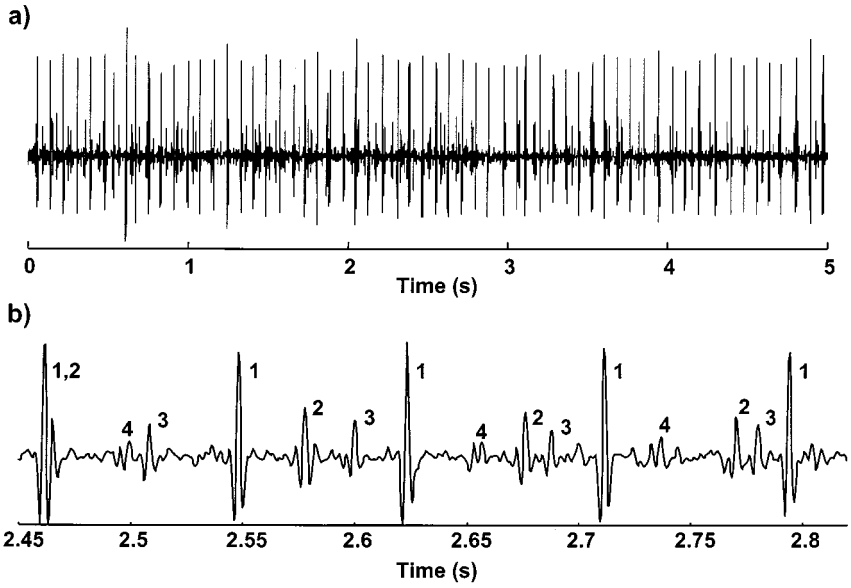

Fig. 7. (a) Surface EMG signal detected by a concentric-ring system constituted by a central point electrode and two rings with radii 5 mm and 10 mm from the tibialis anterior muscle at 20% MVC of contraction level. One MU is clearly identifiable from the surface EMG signal and its firing frequency can be estimated. (b) A part of the signal shown in (a). The activity of the MU with largest MUAPs is evident. Moreover, two to three smaller MUs can be identified and their firings detected when superposition with the larger MUAPs is limited. In (b), MUAPs have been manually classified as belonging to four MUs.

first recording prototype (with the matrix and the concentric system on the same support) from the biceps brachii muscle at 30% of the maximal voluntary contraction force (MVC). The signals have been recorded from the distal part of the muscle (between the innervation zone and the distal tendon region). In different trials, the concentric system was placed in the distal or proximal position (that is closer to the tendon region or to the innervation zone). The conclusions drawn were very similar with the concentric system in the two positions, thus, anatomical differences were excluded in the interpretation of the results. From the single-differential signals, it is not possible to identify any MUAP. The branched electrode in the direction transversal to the muscle fibers and the inverse binomial filter of order two improve source separation. The one-ring concentric system produces a signal with characteristics similar to those of the inverse binomial filter of order two, as obtained in simulation, with MUAPs slightly sharper. The two-ring concentric system presents the best source separation.

Fig. 7 shows a signal detected from the tibialis anterior muscle at low force level (20% MVC) with a two-ring system with radii multiple of 5 mm. The system was placed between the most distal innervation zone and the distal tendon region. From the signal shown, it is possible to reliably detect the firing pattern of at least one MU. The activity of 2–3 smaller MUs can also be detected when moderate superposition with the larger MUAPs is present.

Fig. 8 shows signals detected with the second recording system (concentric rings with radii multiple of 2.5 mm) from the abductor digiti minimi muscle at high contraction level (80% MVC). The system was placed between the innervation zone and the distal tendon region. The signals detected with one-ring, two-ring, and three-ring recording technique are shown. It is still possible to identify single MUAPs even by the analysis of a single EMG channel. The three systems act in a similar way but, as expected theoretically, the selectivity increases with the number of rings. The improvement obtained with more than three rings is negligible (thus, the signals obtained with four and five rings are not shown).

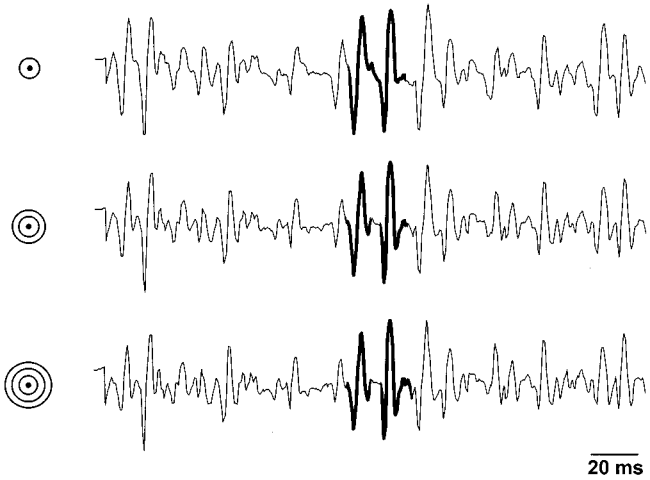

Fig. 8. Examples of signals detected by the one-ring, two-ring, and three-ring concentric systems from the abductor digiti minimi muscle during a high force voluntary contraction (80% MVC). The radii are multiple of 2.5 mm. Note the higher spatial selectivity obtained by increasing the number of rings reflected in the smaller duration of the MUAPs and better separation of the sources (e.g. thick llines).

IV. Discussion and Conclusion

Bhullar *et al.* [2] proposed a system comprised of a point electrode and a ring electrode with opposite and balanced weights and derived the transfer function of this system with a 1-D approximation. Disselhorst *et al.* [8] proposed 2-D grid systems whose transfer function approximated an isotropic function as improvement of the normal double-differential filter [34], [35]

and indicated in the isotropy property of the spatial transfer function an important characteristic for spatial selectivity. In [8], the inverse binomial filter of order two was indicated as that with the best performance among a number of 2-D systems derived from the image processing field. Spatial isotropy implies that the signal detected is independent of the direction of propagation of the sources [13], [15], a feature important in EMG recordings since the orientation of the fibers is not always easy to identify and in some cases may be different in the same muscle for different MUs. Amplitude and spectral content of the EMG signal may be very sensitive to electrode misalignment with conventional EMG recording systems [15]. The proposed concentric-ring systems are isotropic, thus, they do not present the problem of electrode orientation. Spatial isotropy is not directly related to spatial selectivity, since the latter depends mainly on the rejection of the low frequencies from the detected signal.

The systems proposed by Bhullar *et al.* [2] and by the group led by Rau [8], [34], [35] have not been compared in the past from the theoretical point of view since the only available expression of the transfer function of the single-differential concentric system was a 1-D approximation. Thus the theoretical basis for a comparison was lacking. We have presented in this paper the derivation of the transfer function of electrode systems comprised of nonpoint electrodes using the same analytical tools as those used for the characterization of 2-D grid electrode systems. Equation (11) is the counterpart of the delta function modeling the impulse response of point electrodes. Two-dimensional grid electrode systems can thus, be compared with the proposed new systems (Fig. 4). In particular, the derivation of this work indicates that the system proposed by Bhullar *et al.* [2] is theoretically more selective than a number of 1-D and 2-D systems (Figs. 3 and 4). Moreover, it is isotropic while grid systems have a periodic transfer function which can be considered isotropic just for small spatial frequencies. We have also shown that the spatial selectivity can be theoretically improved by increasing the number of rings (Figs. 5, 6, and 8). From the few simulated signals presented, branched electrode configuration with poles transversal to the muscle fibers was very selective in the transversal direction, confirming previous results [19], [23], [41]. However, for source separation the proposed multiple- concentric systems appeared superior, especially for the higher selectivity in the direction parallel to the muscle fibers (Figs. 5 and 6).

It is worthwhile to note that this theoretical study has been performed by assuming a very simple model for EMG generation and detection. In particular, the modelization of electrodes with physical dimensions was done assuming no influence of the electrodes on the potential field distribution. The same simplification was also used by Bhullar *et al.* [2] and Kossev *et al.* [23] but it may be a rather strong approximation in case of concentric-ring systems since the field inside each ring can be changed by the presence of the ring itself [29], [40]. The exact computation of the lead field of ring electrodes is not possible analytically even for a very simple volume conductor model [40]. It may be that the small differences observed between systems with different number of rings when the rings are close (Fig. 8) are due to this approximation which in the future should be numerically investigated. The larger difference in source separation between the one- and two-ring concentric systems when the rings are separated by 5 mm (Fig. 5) with respect to the case in which the rings are separated by 2.5 mm (Fig. 8) should be noted. Moreover, all the theoretical derivation for concentric-ring system design is based on the assumption that MUAPs are traveling without generation or extinction of the intracellular action potential, thus, spatial selectivity has to be intended as selectivity with respect to the traveling components of the EMG signal and not, for example, in the sense of higher reduction of crosstalk. The present derivation does not allow any theoretical comparison related to nontraveling potential rejection. Reduction of nontraveling components depends on the spatial arrangements of the recording electrodes and volume conductor anisotropy [15] [see, also Fig. 5(f)]. However, we were not focusing on crosstalk in this study but on the analysis of single MU activities and in this respect we have shown that the proposed systems present some advantages with respect to previously available systems both from the theoretical point of view (within the approximations introduced) and by experimental evidence.

The presented results are promising for the issues of decomposition of the surface EMG signal into its constituent MUAP trains (Fig. 7) and, in general, of the investigation of single MU properties. In addition, the proposed systems can be easily used in clinical environments and not only in research activities since their application is not particularly more complex than traditional systems such as the single- or double-differential recordings.

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Dario Farina (M'01) graduated *summa cum laude* in electronics engineering from Politecnico di Torino, Torino, Italy. Since 1999 he has been working toward the Ph.D. degree at the Department of Electronics of Politecnico di Torino within a co-tutelle program with the Ecole Centrale de Nantes, Nantes, France, and is involved in teaching activities in electronics and mathematics at Politecnico di Torino.

During 1998 he was a Fellow of the Laboratory for Neuromuscular System Engineering in Torino, Italy. His main interests are in the areas of signal processing applied to biomedical signals, modeling of biological systems and basic and applied physiology of the neuromuscular system.

Mr. Farina is a Registered Professional Engineer.

Corrado Cescon graduated in electronics engineering from Politecnico di Torino, Torino, Italy, in December 2000.

He is currently a Fellow of the Laboratory for Neuromuscular System Engineering in Torino, Italy. His research focuses on signal processing applied to biomedical signals, spatial filtering techniques in surface electromyography, mechanomyography, and measurement of skin impedance variations during acoustic stimulation.